Proposed Plan of Study

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Whose Contribution Counts? AI Mediation and Collective Authorship in Digital Improvisation

In collaborative creative tools, a generative model reinterprets what participants contribute: one person's phrase becomes a sound or a moving form in a shared scene, and its outputs may become more visible than the contributions that prompted them. My doctoral research asks how participants, facilitators and a generative-AI interface shape whether a person's contribution is recognised, carries creative authority, and is credited in a collective work. I will examine whether technical attribution, the system's record of who supplied what, and social recognition among collaborators come apart. This is a question of AI-mediated communication. I study AI not as a creative partner but as a medium that translates one person's offer to others, and ask whose interpretations gain authority and how authorship is negotiated.

My objects of study are small-group digital improvisation workshops with adult improvising artists and creative practitioners, and the shared audiovisual environment I will build for them. Participants contribute short phrases. An existing language model proposes transformations into light, form and sound, and contributors or groups can accept, revise, reject or undo them. Anyone can take part without using AI.

I plan to use the program's research-creation option. The environment, the improvised works made with it, and a set of improvisation scores tested in the workshops form the creative component. The methods are:

- scoping interviews with improvisers and facilitators;
- co-design with practitioners;
- repeated workshops (provisionally three small groups, four sessions each) comparing contributor-led, group-led and human-mediated arrangements;
- reflection interviews in which participants review video of their own sessions.

I will analyse episodes through talk, action and system records, and interviews thematically. The framework is Lucy Suchman's account of how agency is figured at the human–machine interface, complemented by improvisation studies' attention to listening and response.

The project fits Technology in Practice, with Media & Culture as a minor. I hope to work with Dr. Rebecca Caines, whose research on improvisation as a research methodology, socially responsive creative technologies, and generative AI as a critical improvisation practice fits this approach.

My background may appear distant from communication and culture. I hold a Bachelor of Architecture from Tabriz Islamic Art University and a Master of Landscape Architecture from Istanbul Technical University. There I studied perception of virtual environments, work published at eCAADe 2023. I am completing a thesis-based master's degree in Interactive Arts and Technology at Simon Fraser University, where I built and tested multiplayer Unity systems, including ambient cues that let remote teammates perceive one another's heart-rate state without emotion labels (ACM ISS 2026 poster; another paper under revision). Participants made acceptance of that display conditional on controlling who could see it, which led me to ask who controls

how a shared system interprets people. I have conducted ethics-approved mixed-methods studies, worked as a teaching assistant in animation and interactive media, and served on SFU's Senate Graduate Studies Committee.